

INDUSTRIAL TEST SPECIFICATION

MDIO-INPUT1 – 16-Channel 24V Digital Input Module

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This document defines all acceptance tests required for the production release of the MDIO-INPUT1 PCB assembly.

Document Revision History

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A	2026-06-30	HW Engineering	Initial release

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1 Scope and Purpose

This specification defines the mandatory incoming inspection, electrical acceptance tests, and LED chromaticity/photometric tests required to validate every production unit of the MDIO-INPUT1 PCB assembly prior to shipment or integration into end systems.

The MDIO-INPUT1 is a 16-channel, 24 V DC digital input module intended for use in industrial automation and process-control environments. Its primary function is to receive field-side digital signals in the 24 V range, provide galvanic isolation via quad optocouplers, and report the logic states over a standard I²C bus using a PCF8575 16-bit port expander. Two physical I²C connection options (pin header and flat cable/IDC) are provided.

All tests documented herein apply to Class 2 performance per IPC-A-610 unless otherwise stated. Tests are mandatory and must be completed in the sequence given. A unit that fails any test is quarantined and dispositioned before it may proceed to subsequent tests.

2 Referenced Standards and Documents

Standard / Document	Title / Scope
IEC 61131-2:2017	Programmable controllers – Equipment requirements and tests (digital input Types 1/2/3)
IPC-A-610J:2024	Acceptability of Electronic Assemblies (visual workmanship, Class 2)
J-STD-001H:2020	Requirements for Soldering Electrical and Electronic Assemblies
IEC 61010-1:2010	Safety requirements for electrical equipment for measurement, control (clearance/creepage)
IEC 60068-2-1/2/14	Environmental testing – Cold, dry heat, thermal shock (reference only)
CIE Publication 015:2018	Colorimetry – CIE 1931 (x,y) chromaticity coordinate system
IEC 60073:2002	Basic and safety principles – Coding for indicators and actuators (LED color meanings)
IEC 62368-1:2018	Audio/video, IT & comm equipment – Safety requirements
LTV-844 Datasheet	Lite-On Technology, Photocoupler 4-ch 5000 Vrms DIP-16 (DS-70-96-0013)
PCF8575 Datasheet	Texas Instruments, 16-bit I ² C/SMBus I/O expander (SCPS139D)
1N5817 Datasheet	Vishay, 1A Schottky Barrier Rectifier

3 Device Under Test (DUT) Description

The MDIO-INPUT1 PCB is a single-sided SMD/THT mixed-technology assembly. The table below lists all key components extracted from the verified netlist:

Ref	Value / Part	Package	Function
U1	PCF8575DBR	SSOP-24	16-bit I ² C I/O expander; base I ² C address 0x20 – 0x27 set by A0/A1/A2
U2, U3	LTV-844	DIP-16	Quad phototransistor optocoupler, 5000 Vrms isolation; channels 1–8
U4, U5	LTV-844	DIP-16	Quad phototransistor optocoupler, 5000 Vrms isolation; channels 9–16

D1–D16	LED (0805)	SMD	Per-channel status indicators. Required color: YELLOW/AMBER (λ_d 585–595 nm)
D17	1N5817	DO-41	Schottky diode; reverse-polarity protection on VDD supply rail
D18	LED (0805)	SMD	Power-on indicator. Required color: GREEN (λ_d 520–535 nm)
R1–R16	2 k Ω	SMD 0805	Series current-limiting resistors for 24 V optocoupler inputs (IEC 61131-2 compliance)
R17–R32	330 Ω	SMD 0805	Collector pull-up / LED current-setting resistors for status LEDs D1–D16
R33	330 Ω	SMD 0805	Current-setting resistor for power indicator LED D18
C1	100 nF	SMD 0805	VDD bypass/decoupling capacitor
C2	10 μ F	SMD 0805	VDD bulk decoupling capacitor
J1	8-way terminal block	PT-1.5 3.5 mm	24 V field input channels 1–8 (OPTO_IN1–IN8)
J10	9-way terminal block	PT-1.5 3.5 mm	24 V field input channels 9–16 + COM (field ground return)
J9	2-way terminal block	PT-1.5 3.5 mm	5 V DC power input (VDD+ and GND)
J6	4-pin header (right-angle)	2.54 mm	I ² C bus connector option A – Pin header (VDD, SDA, SCL, GND)
J8	4-pin socket (right-angle)	2.54 mm	I ² C bus connector option B – Pin socket (VDD, SDA, SCL, GND)
J4	2×4 IDC header	2.54 mm	I ² C bus connector option C – flat ribbon cable (VDD, GND×4, SCL, SDA, NC)
J3, J5, J7	3-pin jumper header	2.54 mm	Address configuration jumpers ADD0/ADD1/ADD2

3.1 Signal Path Architecture

Each of the 16 digital input channels follows an identical signal path:

- (1) 24 V field signal enters via J1 (ch. 1–8) or J10 (ch. 9–16) and returns to field via the COM terminal (J10 pin 9).
- (2) A 2 k Ω series resistor (R1–R16) limits the optocoupler LED forward current to ≈ 11.4 mA at 24 V nominal, complying with IEC 61131-2 Type 1.
- (3) The optocoupler LED (LTV-844, one of four channels per package) converts the field signal to an optically isolated phototransistor output on the 5 V logic side.
- (4) A 330 Ω pull-up resistor (R17–R32) connects VDD through a status LED (D1–D16) to the phototransistor collector. When the channel is active the transistor conducts, illuminating the LED and pulling the U1 port pin LOW.
- (5) U1 (PCF8575) reports all 16 channel states over I²C. Active input = logic 0 on the corresponding port pin (P00–P07 for channels 1–8; P10–P17 for channels 9–16).

3.2 I²C Address Map

The I²C 7-bit address is: 0b010 0 A2 A1 A0, i.e. base 0x20 plus the binary value of the three address jumpers (J7/J5/J3).

ADD2 (J7)	ADD1 (J5)	ADD0 (J3)	7-bit Address (hex)	Notes
GND	GND	GND	0x20	Factory default
GND	GND	VDD	0x21	
GND	VDD	GND	0x22	
GND	VDD	VDD	0x23	
VDD	GND	GND	0x24	
VDD	GND	VDD	0x25	
VDD	VDD	GND	0x26	
VDD	VDD	VDD	0x27	Max address

4 Test Equipment

The following calibrated instruments are required. All equipment must be within its calibration interval and traceable to national/international measurement standards (NIST or equivalent).

ID	Instrument	Min. Specification	Used In
TE-01	Bench DC Power Supply	0–30 V, 0–3 A, voltage accuracy ± 0.1 V	ET-01, ET-02, ET-06, ET-07, ET-08
TE-02	Digital Multimeter (DMM)	Resolution $\geq 4\frac{1}{2}$ digits; voltage accuracy $\pm 0.1\%$; current accuracy $\pm 0.5\%$	ET-01 through ET-04, ET-08
TE-03	Dielectric Withstand (Hi-Pot) Tester	AC 0–3000 Vrms, leakage current resolution ≤ 0.1 mA; compliant with IEC 61010-1	ET-03
TE-04	Insulation Resistance Tester	500 V DC / 1000 V DC output; resistance range ≥ 10 G Ω	ET-04
TE-05	I ² C Protocol Analyser / Bus Master	100 kHz / 400 kHz capable; read/write access; address 0x20–0x27	ET-05, ET-09, ET-10
TE-06	5 V DC Regulated Supply	± 50 mV regulation; current limit ≥ 0.5 A	ET-02, ET-05 through ET-10, LT series
TE-07	Tristimulus Colorimeter or Spectroradiometer	CIE 1931 (x,y) measurement; wavelength range 380–780 nm; accuracy $\Delta(x,y) \leq 0.002$	LT-01, LT-02
TE-08	Luminance / Luminous-Intensity Meter	Range: 0.1 mcd – 1000 mcd; accuracy $\pm 10\%$	LT-03
TE-09	Precision Decade Resistance Box	10 Ω – 100 k Ω ; accuracy $\pm 0.1\%$	ET-06, ET-07
TE-10	Anti-Static Workstation & Wrist Strap	Per IEC 61340-5-1; resistance to ground 10^6 – 10^9 Ω	All tests

TE-11	Oscilloscope (optional)	Bandwidth ≥ 10 MHz; 2 channels with probe	ET-05 (signal integrity)
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5 Safety Requirements

The following safety precautions are MANDATORY. Failure to observe them may result in personal injury or destruction of the DUT.

- ESD protection: Handle DUTs only at an anti-static workstation. Wear a grounded wrist strap at all times (TE-10). PCF8575 and SMD passive components are susceptible to ESD damage.
- Hi-Pot safety: The dielectric withstand test (ET-03) applies up to 1500 Vrms. Keep hands clear of the DUT and all leads during the test. Use safety-rated test probes with shrouded contacts (IEC 61010 Cat II minimum). Ensure the hi-pot tester (TE-03) has automatic discharge on completion.
- 24 V supply: When applying 24 V field voltage (ET-06, ET-07, ET-08, ET-09), ensure the power supply current limit is set to ≤ 200 mA before connection to prevent damage in the event of a short circuit.
- Power sequencing: Apply 5 V logic supply (TE-06) to J9 before applying 24 V field supply (TE-01) to J1/J10 inputs. Remove 24 V before removing 5 V supply.
- Isolation barrier integrity: Never connect the 24 V field ground (COM) to the 5 V logic GND during normal functional testing. These domains are isolated by the optocouplers. Bridging them invalidates hi-pot test results and risks damaging isolation.
- Optical measurement: When performing LED chromaticity tests, do not look directly into lit LEDs. Use the colorimeter/spectroradiometer contact adapter. Ambient light in the measurement area must be < 1 lux during photometric measurements.

6 Environmental Conditions

All tests shall be performed under the following ambient conditions unless a specific test states otherwise:

Parameter	Requirement
Ambient Temperature	20 °C $\pm$ 5 °C
Relative Humidity	45 % – 75 % RH (non-condensing)
Atmospheric Pressure	860 hPa – 1060 hPa
Magnetic / RF environment	No known EMI sources within 1 m of DUT during hi-pot or I ² C tests
Warm-up time	DUT must be powered at 5 V for ≥ 2 minutes before I ² C and LED tests

7 Pre-Test Visual Inspection (VI-01)

Visual inspection shall be performed per IPC-A-610J Class 2 criteria using $\times 3$ to $\times 10$ magnification. This inspection must be completed and recorded BEFORE any powered test is applied to the DUT.

Item	Inspection Point	Accept Criterion (IPC-A-610J Cl.2)
VI-01a	PCB substrate	No delamination, measing, crazing, blistering, or burns on any laminate layer. Board edge must be free of chipping greater than 0.5 mm into the copper/trace area.

VI-01b	SMD solder joints (R1–R33, C1–C2, D1–D16, D18, U1)	Solder fillet present on $\geq 75\%$ of pad length (minimum toe fillet). No bridging, cold joints, or solder balls. Wetting angle $\leq 90^\circ$. No lifted pads. No visible voids exceeding 25 % of joint area.
VI-01c	Through-hole solder joints (U2–U5, D17)	100 % hole fill (solder visible on component side). Lead protrusion 0.5 mm – 2.5 mm after trim. Concave fillet on solder side. No rough or grainy solder.
VI-01d	Component presence & orientation	All 74 components (per BOM) present and populated. LEDs D1–D18 polarity correct (cathode marking / dot toward GND pad per silkscreen). Electrolytic-like C2 polarity marking correct.
VI-01e	Component value verification (sampling)	Random sample of ≥ 5 SMD resistors: confirm resistor code matches BOM ($2k\Omega = 2001$; $330\Omega = 331$). LED package type 0805 confirmed by footprint measurement.
VI-01f	Terminal blocks J1, J9, J10	Screws present and not pre-torqued. Connectors seated flush with PCB. No cracked or chipped housing. All wire-entry holes clear of solder.
VI-01g	Pin headers / sockets J3, J5, J6, J7, J8 and IDC J4	Pins straight and not bent. Socket J8 contacts not spread or deformed. IDC J4 strain-relief bar present and fully engaged.
VI-01h	Silkscreen and marking	Board reference silkscreen legible. Document/revision marking visible. Channel labels CH1–CH16 on J1/J10 legible.
VI-01i	Cleanliness	No visible flux residue, contamination, solder balls free on board, or foreign material. No conductive particles visible with naked eye.

Disposition: Any defect meeting IPC-A-610J Class 2 'Defect' condition is cause for immediate rejection. 'Process Indicator' conditions must be logged and reviewed by the Engineering team within 24 hours.

8 Electrical Tests

ET-01 – Power Supply & Reverse-Polarity Protection

Objective	Verify the power input circuit accepts 5 V DC in correct polarity, that VDD is correctly distributed on board, and that reverse polarity is blocked by D17.
References	D17 (1N5817), J9, C1, C2; Board netlist: J9 pin2 → D17 anode → D17 cathode → VDD rail

Step	Action / Stimulus	Measurement / Observation	Accept Criterion
1	Set TE-01 to +5.0 V, current limit 200 mA. Connect (+) to J9 pin 2, (–) to J9 pin 1 (GND). Do NOT connect I ² C or field inputs.	Measure VDD at C1 pad 2 vs. C1 pad 1 (GND).	$4.7\text{ V} \leq \text{VDD} \leq 5.3\text{ V}$. Supply current $\leq 80\text{ mA}$ (idle).
2	Record supply current from TE-01 display.	Current reading.	Idle current: $5\text{ mA} \leq I_{\text{idle}} \leq 80\text{ mA}$ (D18 lit + PCF8575 quiescent).
3	Remove power. Reverse polarity: connect (–) to J9 pin 2, (+) to J9 pin 1. Apply 5 V.	Measure VDD at C1 pad 2. Observe power indicator LED D18.	$\text{VDD} < 0.5\text{ V}$ (D17 blocking). D18 must be OFF. Supply current $< 2\text{ mA}$

			(leakage only). No damage.
4	Remove power. Restore correct polarity for remaining tests.	Visual confirmation of correct wiring.	Correct polarity confirmed.

ET-02 – VDD Rail Voltage Verification

Objective	Confirm VDD rail accuracy across the PCB under loaded conditions representative of all channels active.
References	PCF8575 VDD: 2.5 V–5.5 V per datasheet; nominal operating point 5.0 V

Step	Action / Stimulus	Measurement / Observation	Accept Criterion
1	Power DUT with 5.0 V at J9 (correct polarity). All field inputs disconnected.	Measure VDD at U1 pin 24 vs. GND (U1 pin 12 or 24).	$4.75\text{ V} \leq \text{VDD} \leq 5.25\text{ V}$.
2	Simultaneously apply a 24 V signal to all 16 input channels (all optocouplers conducting). Measure VDD.	Measure VDD at U1 pin 24 under full load.	$\text{VDD} \geq 4.65\text{ V}$ ($\leq 350\text{ mV}$ drop from TE-06 setting).
3	Record min VDD seen during step 2.	Min VDD reading.	Min VDD $\geq 4.65\text{ V}$.

ET-03 – Isolation / Hi-Pot (Dielectric Withstand Test)

Objective	Verify that the galvanic isolation barrier between the 24 V field side and 5 V logic side can withstand 1500 Vrms AC for 1 second without breakdown, consistent with the LTV-844 5000 Vrms rated isolation voltage (production test at 30 % of rated).
References	LTV-844 V _{iso} = 5000 Vrms; IEC 61010-1; production hi-pot practice

WARNING: This test uses hazardous voltages. Follow all safety precautions in Section 5. Confirm the hi-pot tester (TE-03) is in calibration and has a working ground-fault interrupt.

Step	Action / Stimulus	Measurement / Observation	Accept Criterion
1	Disconnect all external connections from J1, J9, J10, J6, J8, J4 (DUT completely isolated).	Visual confirmation.	All external connectors free.
2	Short all field-side terminals together: J1 pins 1–8, J10 pins 1–9 (including COM). Connect to HV terminal of TE-03.	Confirm short is secure.	Field-side bus formed.
3	Short all logic-side terminals together: J9 pin 1 (GND), J9 pin 2 (VDD), J6 pins 1–4, J8 pins 1–4, J4 pins 1–8. Connect to LV/GND terminal of TE-03.	Confirm short is secure.	Logic-side bus formed.
4	Set TE-03 to AC test mode, voltage ramp: 0 → 1500 Vrms over 5 seconds, hold 1500 Vrms for 1 second, leakage current trip threshold: 10 mA.	Monitor leakage current throughout test.	Leakage current < 10 mA at all times. No arc, no flashover, no trip.

5	Discharge DUT via TE-03 built-in discharge function. Confirm discharge complete before touching DUT.	TE-03 discharge status.	Discharge confirmed (< 30 V residual).
6	Inspect DUT for signs of damage (burn marks, delamination, component damage).	Visual inspection.	No visible damage. DUT appearance unchanged.

ET-04 – Insulation Resistance

Objective	Confirm that the DC insulation resistance between field and logic domains meets minimum requirements after the hi-pot test.
References	IEC 61010-1: minimum insulation resistance typically $\geq 100 \text{ M}\Omega$ at rated test voltage

Step	Action / Stimulus	Measurement / Observation	Accept Criterion
1	Use the same field-side and logic-side bus shorts as ET-03 (or re-create them).	—	—
2	Apply TE-04 at 500 V DC between field-side bus and logic-side bus.	Measure insulation resistance after 60 s stabilisation.	$R_{\text{iso}} \geq 100 \text{ M}\Omega$.
3	Discharge per TE-04 procedure before disconnecting.	Discharge confirmation.	Complete.

ET-05 – I²C Device Detection & Address Configuration

Objective	Verify U1 (PCF8575) responds on the I ² C bus at the expected address for all 8 jumper combinations (J3/J5/J7), and that SDA and SCL signal integrity is acceptable.
References	PCF8575 datasheet; I ² C specification 400 kHz Fast Mode; J3 (ADD0), J5 (ADD1), J7 (ADD2)

Step	Action / Stimulus	Measurement / Observation	Accept Criterion
1	Power DUT at 5 V via J9. Connect I ² C bus master (TE-05) to J6 pins: pin1=VDD, pin2=SDA, pin3=SCL, pin4=GND (or equivalent via J8 socket).	Confirm bus master powered and TE-05 connected.	Bus connected.
2	Set J3, J5, J7 all to GND (address = 0x20). Issue an I ² C address scan at 100 kHz.	Record device detected at 0x20.	ACK received at 0x20. No spurious devices on bus.
3	Perform a 2-byte read from U1 at 0x20. All inputs floating HIGH (PCF8575 internal pull-up).	Read 16-bit port value.	Both bytes = 0xFF (all ports HIGH = no active channel).
4	Change jumper to ADD0=VDD (J3 center to VDD pin). Repeat scan.	Detect device.	ACK received at 0x21. No response at 0x20.

5	Test remaining 6 jumper combinations (0x22 through 0x27). For each, issue scan and verify correct address.	Record detected address for each jumper combination.	Correct address ACK for all 8 combinations. No address collision.
6	With address at 0x20, measure SDA and SCL waveforms on oscilloscope (TE-11) at 400 kHz (Fast Mode).	Waveform measurements: high level (V_{OH}), low level (V_{OL}), rise time (t_r).	$V_{OH} \geq 0.7 \times V_{DD} = 3.5 \text{ V}$; $V_{OL} \leq 0.3 \times V_{DD} = 1.5 \text{ V}$; $t_r \leq 300 \text{ ns}$ (1 k Ω pull-up).
7	Restore J3/J5/J7 to GND (0x20) for remaining tests.	Confirm address 0x20.	0x20 confirmed.

ET-06 – Digital Input OFF-State Threshold (IEC 61131-2 Type 1)

Objective	Verify that every channel reads logic 1 (inactive) when the input voltage is at or below the Type 1 OFF threshold of 5 V DC, confirming the optocoupler does not falsely trigger.
References	IEC 61131-2:2017 Table 7 – Type 1, 24V DC: $V_{OFF}(\text{max}) = 5 \text{ V}$, $I_{IN}(\text{OFF}) \leq 15 \text{ mA}$

Test is performed channel by channel. Connect TE-01 (+5 V) between the channel terminal (J1 pin N or J10 pin N) and COM (J10 pin 9) via TE-09 (set to 0 Ω bypass). Power DUT at 5 V.

Step	Action / Stimulus	Measurement / Observation	Accept Criterion
1	Apply $V_{IN} = 0 \text{ V}$ to CH1 (J1 pin 1 to COM). Read CH1 via I ² C.	P00 bit value in PCF8575 read data.	P00 = 1 (inactive). D1 is OFF.
2	Ramp V_{IN} from 0 V to 5.0 V in 0.5 V steps. At each step read CH1 bit.	Note voltage at which CH1 first reads '0' (active).	CH1 must remain '1' (inactive) at all voltages $\leq 5.0 \text{ V}$ (exclusive). No false trigger.
3	Measure I_{IN} at $V_{IN} = 5 \text{ V}$.	Current reading (TE-02 series ammeter).	$I_{IN}(5\text{V}) \leq 15 \text{ mA}$ (IEC 61131-2 Type 1 requirement).
4	Repeat steps 1–3 for each of the 16 channels (CH2–CH16). Record results.	16 independent measurements.	All 16 channels: P00–P07, P10–P17 remain '1' at $V_{IN} \leq 5 \text{ V}$.

ET-07 – Digital Input ON-State Threshold (IEC 61131-2 Type 1)

Objective	Verify that every channel reads logic 0 (active) when $V_{IN} \geq 15 \text{ V}$, and that it reports active at the nominal 24 V supply.
References	IEC 61131-2:2017 Table 7 – Type 1, 24V DC: $V_{ON}(\text{min}) = 15 \text{ V}$, $V_{nom} = 24 \text{ V}$, $V_{max} = 30 \text{ V}$

Step	Action / Stimulus	Measurement / Observation	Accept Criterion
1	Apply $V_{IN} = 15.0 \text{ V}$ to CH1 (J1 pin 1 to COM). Read CH1 via I ² C.	P00 bit value.	P00 = 0 (active). D1 must be ON (illuminated).

2	Apply $V_{IN} = 24.0\text{ V}$ to CH1. Read CH1.	P00 bit value.	P00 = 0 (active). D1 ON.
3	Apply $V_{IN} = 30.0\text{ V}$ (maximum operating voltage) to CH1. Read CH1. Monitor supply current.	P00 value and I_{IN} reading.	P00 = 0. $I_{IN}(30V) \leq 15\text{ mA}$ ($= (30 - 1.2)/2000 \approx 14.4\text{ mA}$). D1 ON.
4	Repeat steps 1–3 for all 16 channels. Verify each channel independently.	16×3 measurements.	All channels: active (0) at 15 V, 24 V, and 30 V.

ET-08 – Input Current Verification (IEC 61131-2 Type 1 Compliance)

Objective	Measure actual input current at nominal operating conditions (24 V) to confirm compliance with IEC 61131-2 Type 1 current band and verify correct resistor values R1–R16.
References	R1–R16 = 2 k Ω ; LTV-844 $V_F \approx 1.2\text{ V}$; IEC 61131-2 Type 1: $I_{IN}(24V) = 2\text{--}15\text{ mA}$

Expected: $I_{IN} = (24\text{ V} - V_{F_opto}) / R_{series} \approx (24 - 1.2) / 2000 \approx 11.4\text{ mA}$.

Step	Action / Stimulus	Measurement / Observation	Accept Criterion
1	Insert TE-02 (ammeter mode) in series between TE-01 (+24 V) and CH1 terminal. Apply $V_{IN} = 24.0\text{ V}$.	Record I_{IN} for CH1.	$9.0\text{ mA} \leq I_{IN} \leq 14.0\text{ mA}$.
2	Repeat for all 16 channels.	Record I_{IN} per channel.	All 16 channels: $9.0\text{ mA} \leq I_{IN} \leq 14.0\text{ mA}$.
3	Check that all 16 values agree within $\pm 15\%$ of each other.	Calculate max deviation.	Max channel-to-channel deviation $\leq \pm 15\%$ of mean.

ET-09 – Full Channel Functional Test (All 16 Channels)

Objective	Confirm that each channel independently and correctly maps its 24 V field state to the correct bit in the PCF8575 I ² C register, with correct status LED behaviour, using channel walking and all-ON / all-OFF patterns.
References	U1 PCF8575 port map: P00–P07 = CH1–CH8 (byte 0); P10–P17 = CH9–CH16 (byte 1)

Step	Action / Stimulus	Measurement / Observation	Accept Criterion
1	Power DUT, all channels disconnected. Read I ² C. Record baseline.	Baseline read value.	Bytes 0 and 1 both = 0xFF (all inactive). All D1–D16 OFF.
2	Activate CH1 only (24 V to J1 pin 1 / COM). Read I ² C byte 0.	Byte 0 value, D1 status.	Byte 0 = 0xFE (bit 0 = 0). D1 lit. All other LEDs off. Byte 1 = 0xFF.
3	Walking 1 test: successively move the active channel from CH1 to CH16. After each channel activation, verify the correct bit is	$16 \times (\text{I}^2\text{C byte pair, LED status})$.	For channel N: correct bit in correct

	zero and only that bit; verify corresponding LED is lit.		byte = 0. All other 15 bits = 1. Only D_N lit.
4	All-ON test: simultaneously apply 24 V to all 16 channels.	Read I ² C byte pair; visual LED check.	Byte 0 = 0x00, Byte 1 = 0x00. All 16 LEDs D1–D16 illuminated simultaneously.
5	All-OFF: remove 24 V from all channels.	Read I ² C byte pair.	Byte 0 = 0xFF, Byte 1 = 0xFF. All LEDs off.
6	Verify INT pin (U1 pin 1) transitions low upon any channel change. Connect TE-02 to INT and GND.	INT voltage: baseline and during channel event.	INT = VDD (≥ 4.5 V) at baseline. INT ≤ 0.8 V within 50 μ s of channel state change (open-drain with external pull-up on I ² C master).

ET-10 – Dual I²C Connector Verification (J6 Pin Header / J8 Socket / J4 IDC)

Objective	Confirm that both I ² C connection options (J6 pin header and J8 socket) as well as the IDC flat cable connector J4 present identical bus behaviour with correct pin mapping.
References	J6: pins 1=VDD, 2=SDA, 3=SCL, 4=GND; J8: same pinout (socket); J4 (IDC 2x4): pin1=VDD, pin2=GND, pin3=SCL, pin4=GND, pin5=SDA, pin6=GND, pin7=GND, pin8=NC

Step	Action / Stimulus	Measurement / Observation	Accept Criterion
1	Connect TE-05 to J6 (pin header). Apply 5 V. Issue I ² C read at 0x20. Activate CH1 (24 V).	Read byte 0 via J6.	Byte 0 = 0xFE. Correct data via J6.
2	Disconnect TE-05 from J6. Connect to J8 (socket). Repeat read.	Read byte 0 via J8.	Byte 0 = 0xFE. Correct data via J8. SDA/SCL on identical nets.
3	Disconnect TE-05 from J8. Connect flat cable to J4 IDC header. Verify pin 5=SDA, pin 3=SCL, pin 1=VDD, pin 2=GND. Repeat read.	Read byte 0 via J4.	Byte 0 = 0xFE. Correct data via J4 IDC.
4	Measure SDA and SCL signal levels on J4 under 400 kHz bus activity.	V _{OH} and V _{OL} on J4 pins.	V _{OH} ≥ 3.5 V, V _{OL} ≤ 1.5 V (same as ET-05 step 6).

9 LED Chromaticity and Photometric Tests

Per IEC 60073:2002, indicator lamp colors carry normative safety meanings in industrial equipment. The MDIO-INPUT1 assigns specific colors to its LEDs to convey unambiguous status information:

Ref	Color	IEC 60073 Meaning	Application on MDIO-INPUT1
D18	GREEN	Normal / Safe / ON	5 V logic supply present and operational
D1–D16	YELLOW / AMBER	Caution / Active state	24 V field signal present on that input channel

The CIE 1931 (x, y) chromaticity test uses a calibrated colorimeter or spectroradiometer (TE-07) with the probe placed flush against the LED lens at a defined measurement distance (contact or 5 mm $\pm$ 1 mm). All measurements shall be performed in ambient light < 1 lux (darkened test enclosure). Forward current is set at I_F = 10 mA using TE-06 with a series precision resistor calculated per the DUT supply voltage. This current is representative of the in-circuit operating current ($\approx$ 8–10 mA).

9.1 LT-01 – Power Indicator LED D18 Chromaticity (GREEN)

Required Color	GREEN – IEC 60073 code; assigned to 'Normal / Power Present' function
Dominant Wavelength	λ_d = 520 nm – 535 nm
CIE 1931 x	$0.050 \leq x \leq 0.185$
CIE 1931 y	$0.640 \leq y \leq 0.780$
Purity (min)	Excitation purity $\geq$ 90 % (highly saturated, no appreciable white admixture)
Test current I_F	10 mA $\pm$ 0.5 mA DC, applied via precision bench supply and series 300 Ω resistor from 5 V source

Step	Action / Stimulus	Measurement / Observation	Accept Criterion
1	Power TE-06 to 5 V. Connect in series with 300 Ω precision resistor and D18 anode/cathode pads (lift one end of R33 or probe directly). Set current to 10 mA $\pm$ 0.5 mA.	Confirm I_F with TE-02.	I_F = 10 mA $\pm$ 0.5 mA.
2	Place TE-07 probe flush with D18 lens. Darken ambient to < 1 lux. Acquire measurement ($\geq$ 3 readings, average).	Record CIE (x, y) and dominant wavelength λ_d .	x: 0.050–0.185; y: 0.640–0.780; λ_d : 520–535 nm.
3	Verify the measured (x, y) point lies within the GREEN acceptance polygon (points: (0.050, 0.640), (0.185, 0.640), (0.185, 0.780), (0.050, 0.780)).	Plot or compute against acceptance window.	Point within acceptance polygon. PASS.
4	If any reading falls outside the polygon, reject. Log measurement data to test record.	Log data.	See test record form.

9.2 LT-02 – Channel Status LEDs D1–D16 Chromaticity (YELLOW / AMBER)

Required Color	YELLOW / AMBER – IEC 60073 code; assigned to 'Active state / Channel ON' function
Dominant Wavelength	λ_d = 580 nm – 600 nm
CIE 1931 x	$0.400 \leq x \leq 0.540$
CIE 1931 y	$0.400 \leq y \leq 0.560$
Purity (min)	Excitation purity $\geq$ 85 % (high saturation, clearly distinguishable from orange and green)
Test current I_F	10 mA $\pm$ 0.5 mA DC; applied via 5 V supply, 300 Ω series resistor, direct to status LED pads

Sampling plan	100 % test of all 16 channel LEDs (D1–D16). Production volumes < 100 units/batch: full test. ≥ 100 units: AQL 1.5 Level II with Reduced Inspection (7-unit sample minimum).
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Step	Action / Stimulus	Measurement / Observation	Accept Criterion
1	Power DUT at 5 V. Drive CH1 active (24 V on J1 pin 1). D1 lights via normal circuit path at ≈ 8–10 mA.	Confirm D1 is lit.	D1 illuminated.
2	Place TE-07 probe on D1. Acquire CIE (x,y) and λ_d . Record.	CIE (x, y) and λ_d for D1.	x: 0.400–0.540; y: 0.400–0.560; λ_d : 580–600 nm.
3	Repeat step 1–2 for each of D2–D16 by activating each channel in turn. Record each (x, y, λ_d).	16 individual measurements.	All 16 LEDs pass chromaticity window.
4	Check all 16 (x, y) values lie within the YELLOW/AMBER acceptance quadrilateral (points: (0.400, 0.400), (0.540, 0.400), (0.540, 0.560), (0.400, 0.560)).	Pass/fail per LED.	All 16 within polygon. Any failure = batch HOLD, notify Engineering.
5	Optionally compare D1 to D16 chromaticity scatter: max range of x values ≤ 0.060; max range of y values ≤ 0.060.	Chromaticity spread analysis.	Within-batch consistency: $\Delta x \leq 0.060$, $\Delta y \leq 0.060$. This is a process indicator, not a hard fail criterion.

9.3 LT-03 – Luminous Intensity and LED Forward Voltage

Step	Action / Stimulus	Measurement / Observation	Accept Criterion
1	At $I_F = 10$ mA: measure luminous intensity of D18 using TE-08 at probe-to-LED distance 5 mm.	Record I_v (mcd) for D18.	$I_v \geq 2$ mcd at $I_F = 10$ mA (minimum visibility under ambient industrial lighting).
2	At $I_F = 10$ mA: measure luminous intensity of D1, D4, D8, D12, D16 (5 representative channels).	Record I_v (mcd) for 5 LEDs.	All 5: $I_v \geq 2$ mcd at $I_F = 10$ mA.
3	Measure forward voltage V_F of D18 at $I_F = 10$ mA.	Record V_F (V).	$1.8 \text{ V} \leq V_F \leq 3.5 \text{ V}$ (covers all visible LED chemistries).
4	Measure V_F of D1, D4, D8, D12, D16 at $I_F = 10$ mA.	Record V_F per LED.	$1.8 \text{ V} \leq V_F \leq 3.5 \text{ V}$ for all. Channel-to-channel variation ≤ 0.3 V.

10 Final Acceptance and Traceability

A unit is considered a PASS only when ALL of the following conditions are satisfied:

1. VI-01: All visual inspection items received a PASS disposition (no IPC-A-610J Class 2 Defect recorded).

2. ET-01 through ET-10: All electrical tests passed all step criteria without any substitution or re-test (one re-test per step permitted after documented rework, but original and re-test results must both be logged).
3. LT-01 through LT-03: All LED chromaticity coordinates fall within the defined acceptance polygons, and luminous intensity meets minimum values.
4. Test Record Form (Section 11) is fully completed, legible, and signed by the test technician and a second authorized reviewer.

Traceability Requirement	Detail
Unit serialisation	Each DUT must carry a unique serial number (laser-etched or permanent label, min 6 characters alphanumeric) affixed to the PCB before testing begins.
Calibration status	All test equipment IDs and calibration expiry dates must be recorded on the Test Record Form prior to commencing tests.
Non-conformances	Failed units must be tagged with a red HOLD label, assigned an NCR (Non-Conformance Report) number, and stored in a quarantine location. They must not be re-inserted into the production flow without a written Engineering disposition.
Record retention	Completed test records shall be retained for a minimum of 10 years per IEC 61010-1 product liability recommendations.

11 Test Record Form

Unit Serial No.		PCB Part No.	MDIO-INPUT1
Test Date		Test Spec Rev.	A
Technician Name		Reviewer Name	
Ambient Temp (°C)		Humidity (%RH)	

Equipment Used:

TE-ID	Instrument / Model	Calibration Cert No.	Cal Expiry
TE-01			
TE-02			
TE-03			
TE-04			
TE-05			
TE-06			
TE-07			
TE-08			
TE-09			

Test Results:

Test ID	Parameter / Measurement	Measured Value	Accept Criterion	P / F
VI-01	Visual inspection – IPC-A-610J Class 2	—	No Defect conditions	
ET-01a	VDD at C1 (5V applied)	___ V	4.7–5.3 V	
ET-01b	Idle supply current	___ mA	5–80 mA	
ET-01c	VDD at reverse polarity	___ V	< 0.5 V	
ET-02a	VDD at U1 pin 24 (no load)	___ V	4.75–5.25 V	
ET-02b	VDD at U1 pin 24 (all ch active)	___ V	≥ 4.65 V	
ET-03	Hi-Pot 1500 Vrms × 1 s – leakage	___ mA	< 10 mA; no trip	
ET-04	Insulation resistance @ 500 V DC	___ MΩ	≥ 100 MΩ	
ET-05a	I ² C ACK at 0x20	ACK / NACK	ACK	

ET-05b	I ² C ACK all 8 addresses (0x20–0x27)	<i>Pass / Fail</i>	All 8 pass	
ET-05c	SDA/SCL V _{OH} @ 400 kHz	___ V	≥ 3.5 V	
ET-06	OFF threshold – all 16 ch @ 5 V	<i>Pass / Fail</i>	All remain 1 at ≤5V	
ET-07a	ON threshold – all 16 ch @ 15 V	<i>Pass / Fail</i>	All read 0 at 15V	
ET-07b	ON @ 24 V and 30 V – all 16 ch	<i>Pass / Fail</i>	All read 0	
ET-08	I _{IN} @ 24 V – all 16 ch	___ mA (range)	9.0–14.0 mA	
ET-09a	Walking-1 – all 16 channels	<i>Pass / Fail</i>	Correct bit per ch	
ET-09b	All-ON: bytes 0,1 = 0x00	0x___ / 0x___	0x00 / 0x00	
ET-09c	INT pin transitions on change	<i>Pass / Fail</i>	V _{INT} ≤ 0.8 V on event	
ET-10	I ² C data correct via J6, J8, J4	<i>Pass / Fail</i>	All 3 connectors pass	
LT-01	D18 CIE (x,y)	(___,___)	x:0.050–0.185; y:0.640–0.780	
LT-01	D18 dominant wavelength	___ nm	520–535 nm	
LT-02	D1–D16 CIE (x,y) all channels	(range)	x:0.400–0.540; y:0.400–0.560	
LT-02	D1–D16 dominant wavelength	___ nm	580–600 nm	
LT-03a	D18 luminous intensity @ I _F =10mA	___ mcd	≥ 2 mcd	
LT-03b	D1/D4/D8/D12/D16 intensity	___ mcd	≥ 2 mcd each	
LT-03c	D18 forward voltage @ I _F =10mA	___ V	1.8–3.5 V	
LT-03d	D1/D4/D8/D12/D16 V _F	___ V range	1.8–3.5 V; ΔV _F ≤ 0.3 V	

Remarks / NCR References:

Final Disposition:

Role	Print Name	Signature	Date
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Test Technician			
Quality / Reviewer			
Overall Result	☐ PASS ☐ FAIL / HOLD (circle one)		